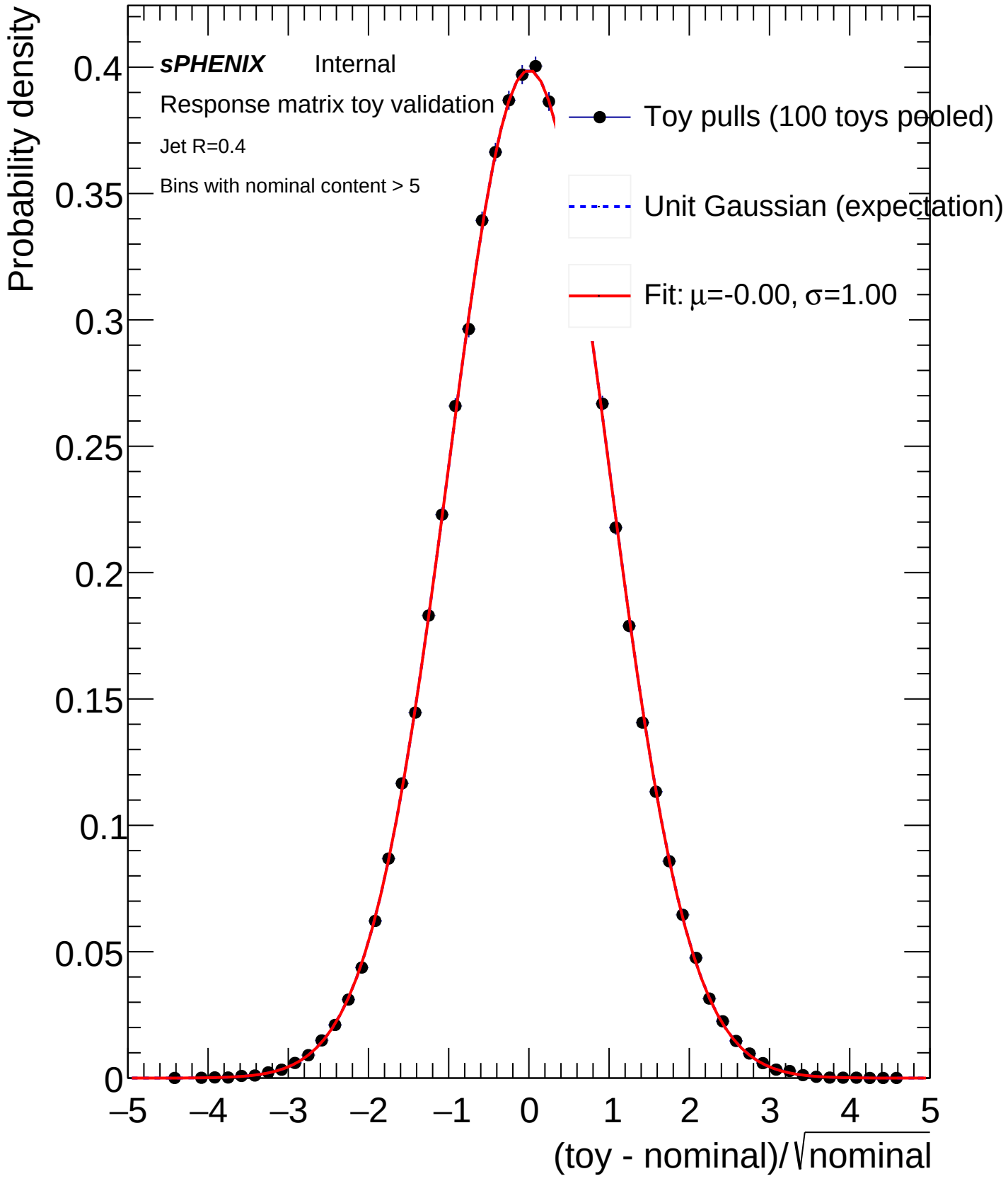
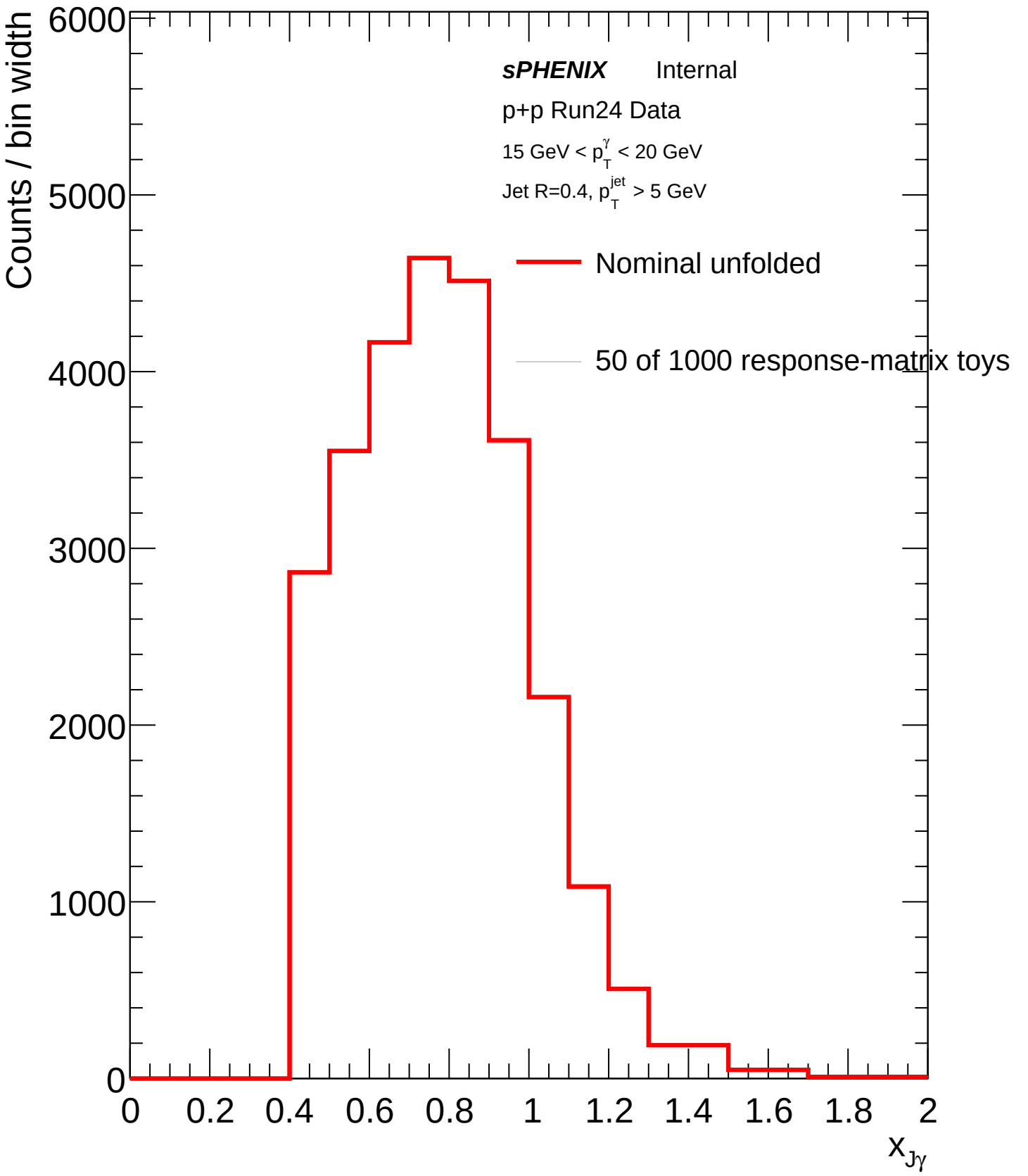
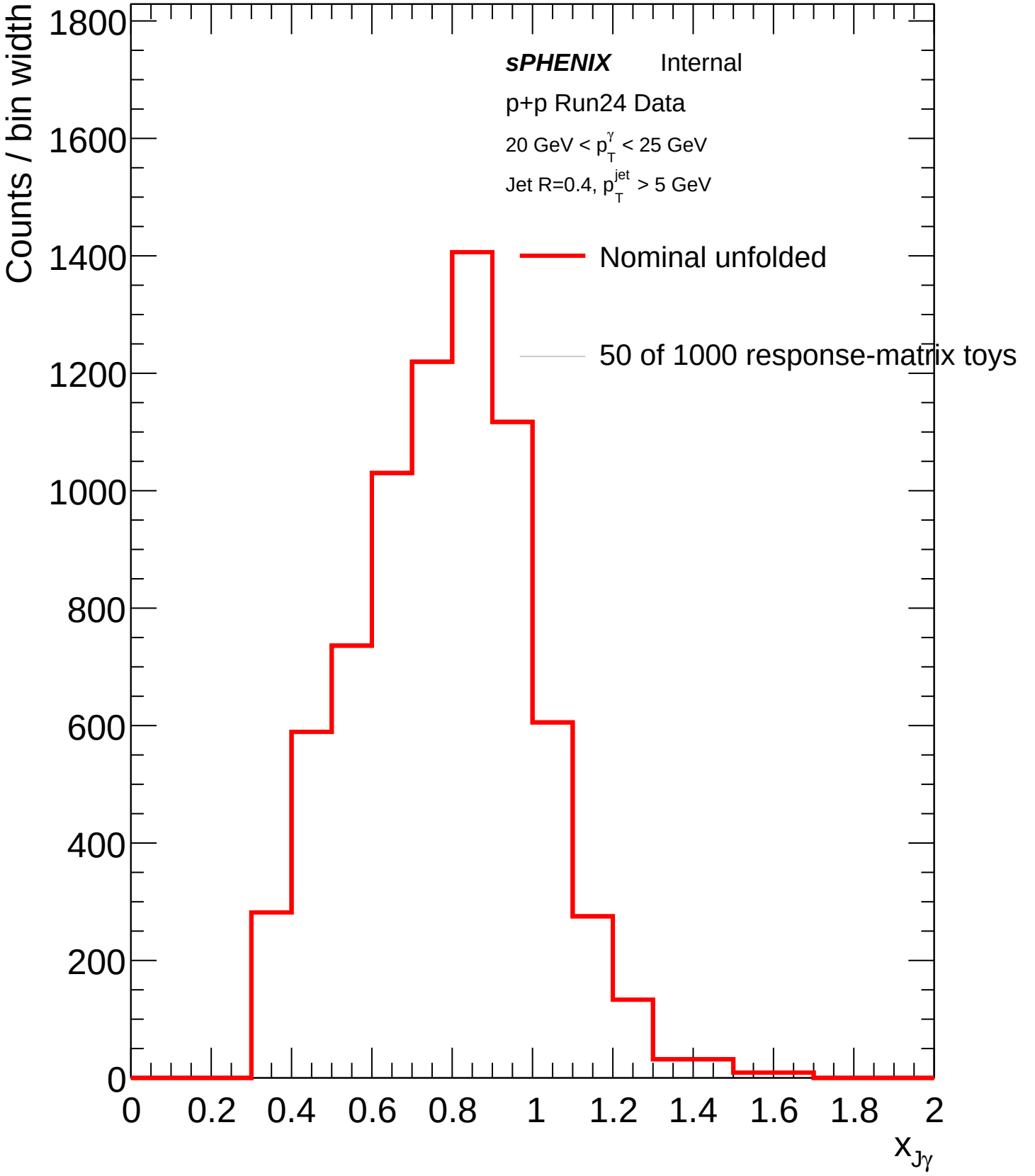
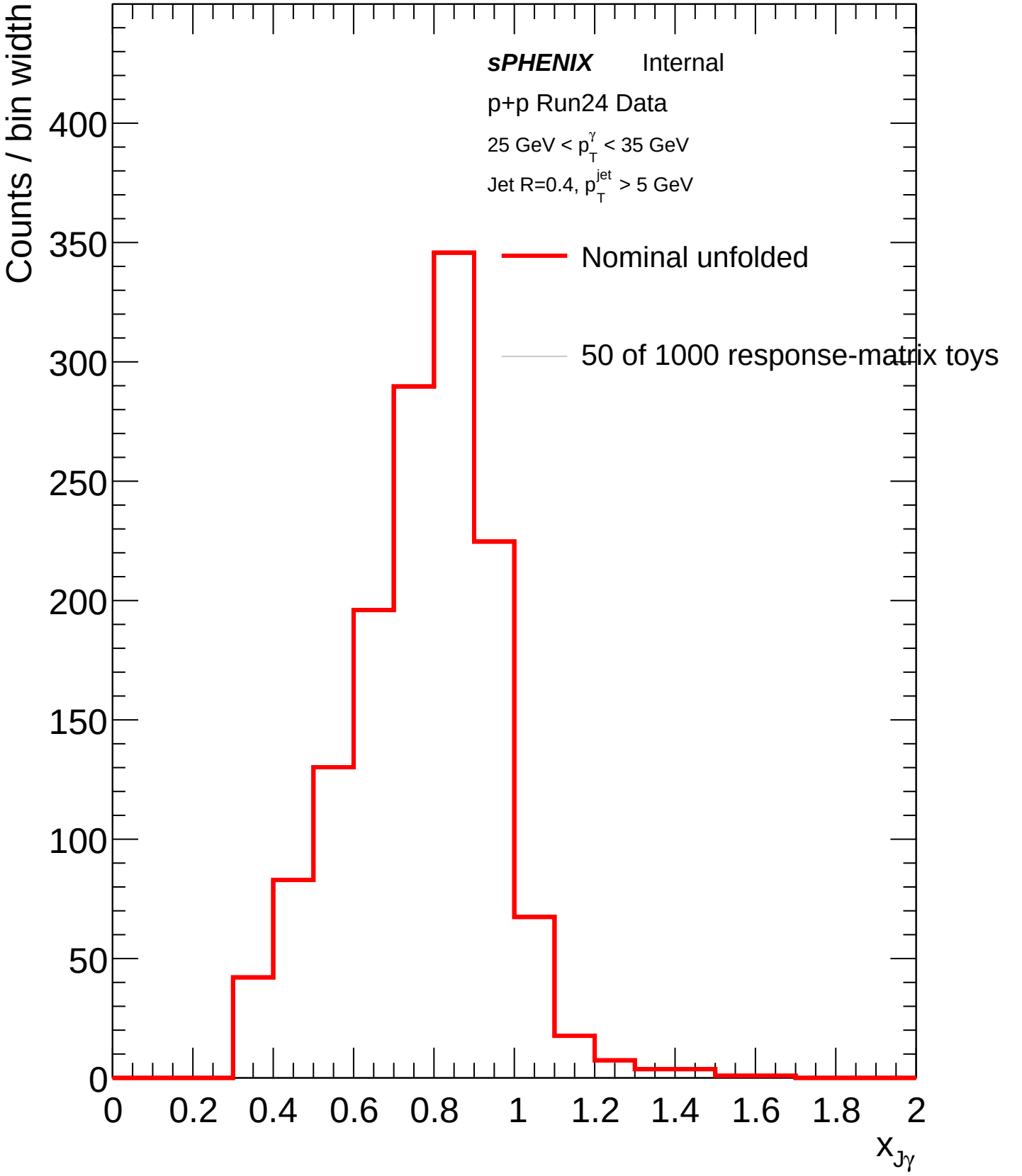
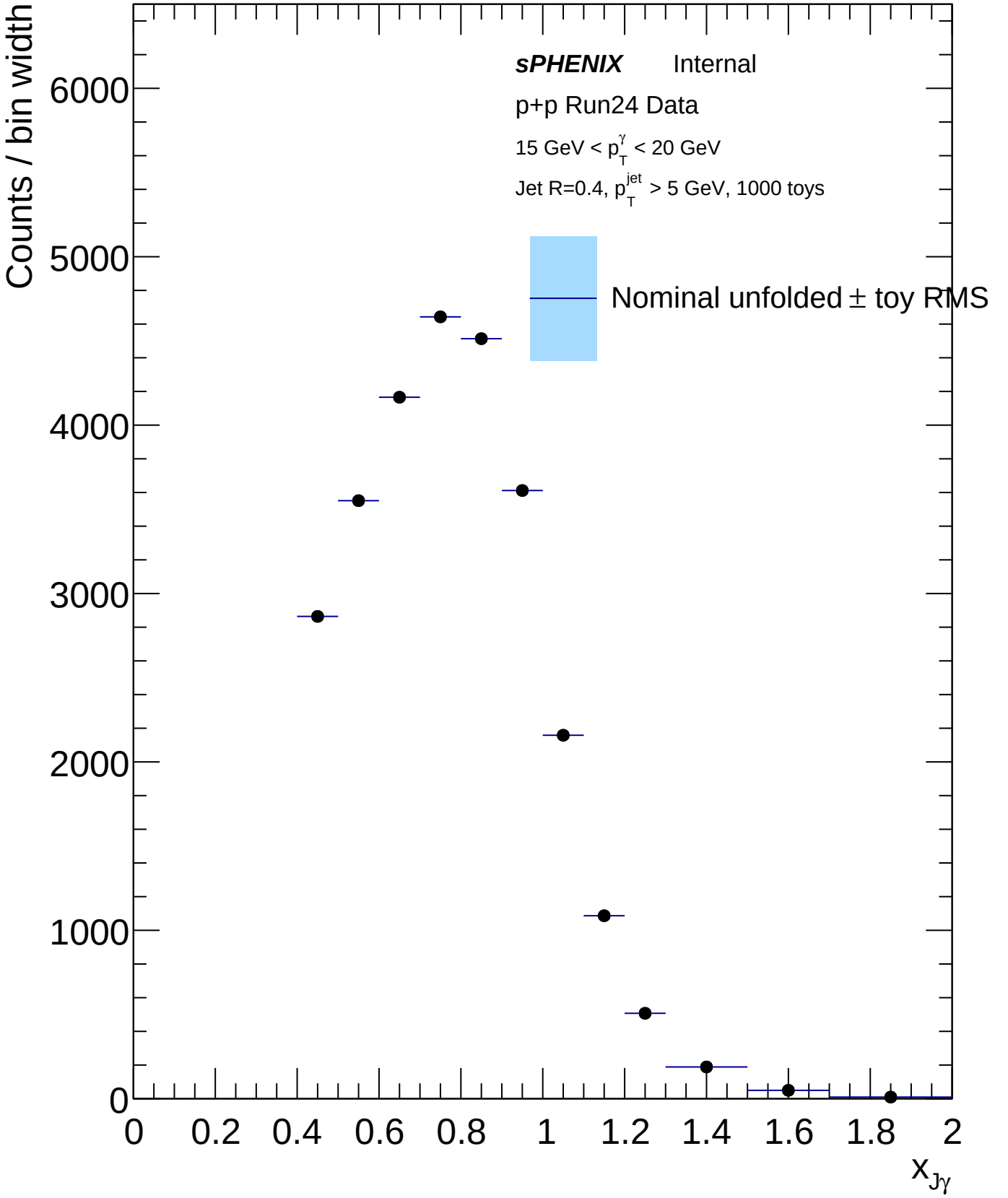
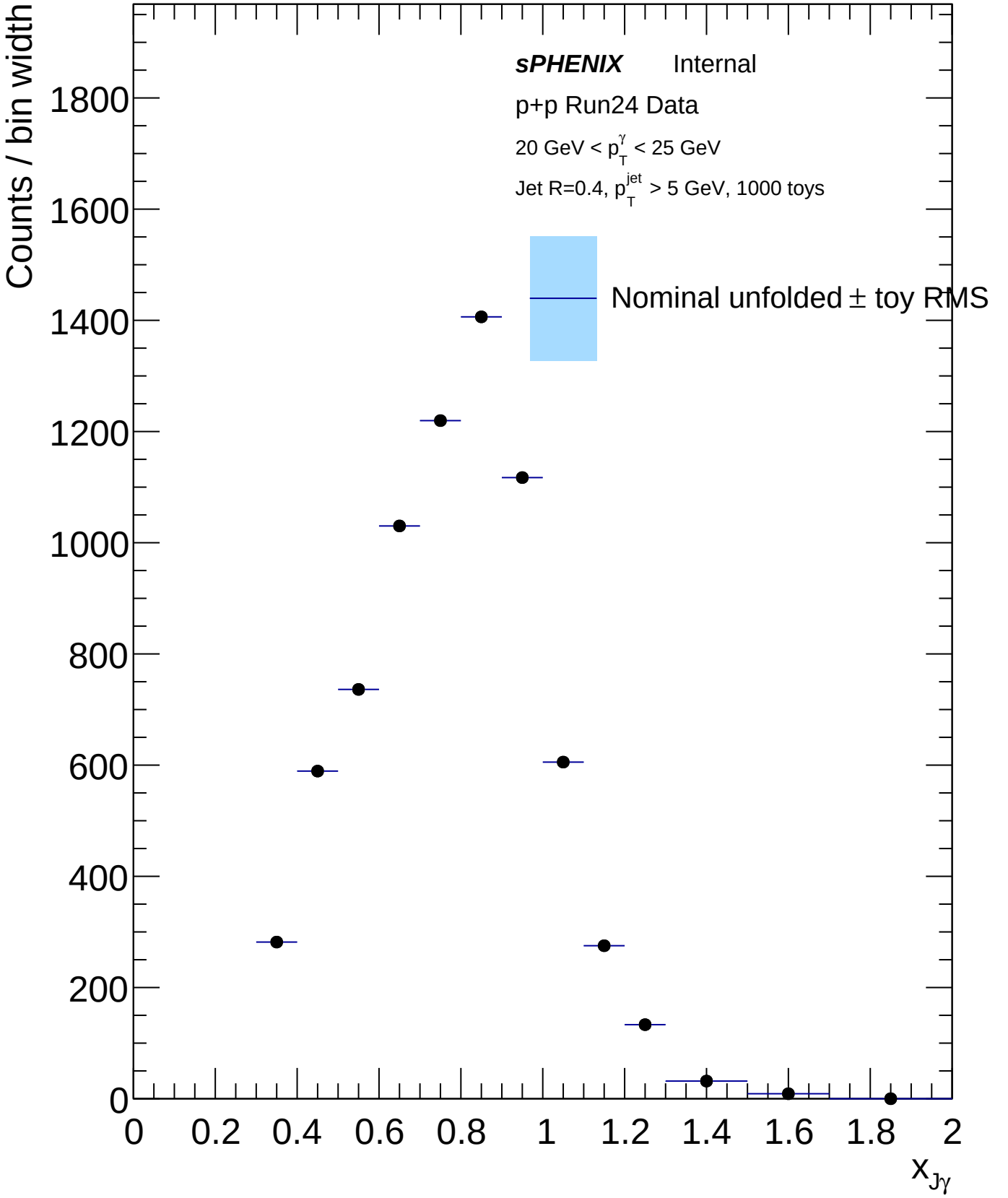

sPHENIX Internal
Response matrix toy validation
Jet R=0.4
Bins with nominal content > 5
Toy pulls (100 toys pooled)
Unit Gaussian (expectation)
Fit: μ=-0.00, σ=1.00
Probability density
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
(toy - nominal)/√nominal

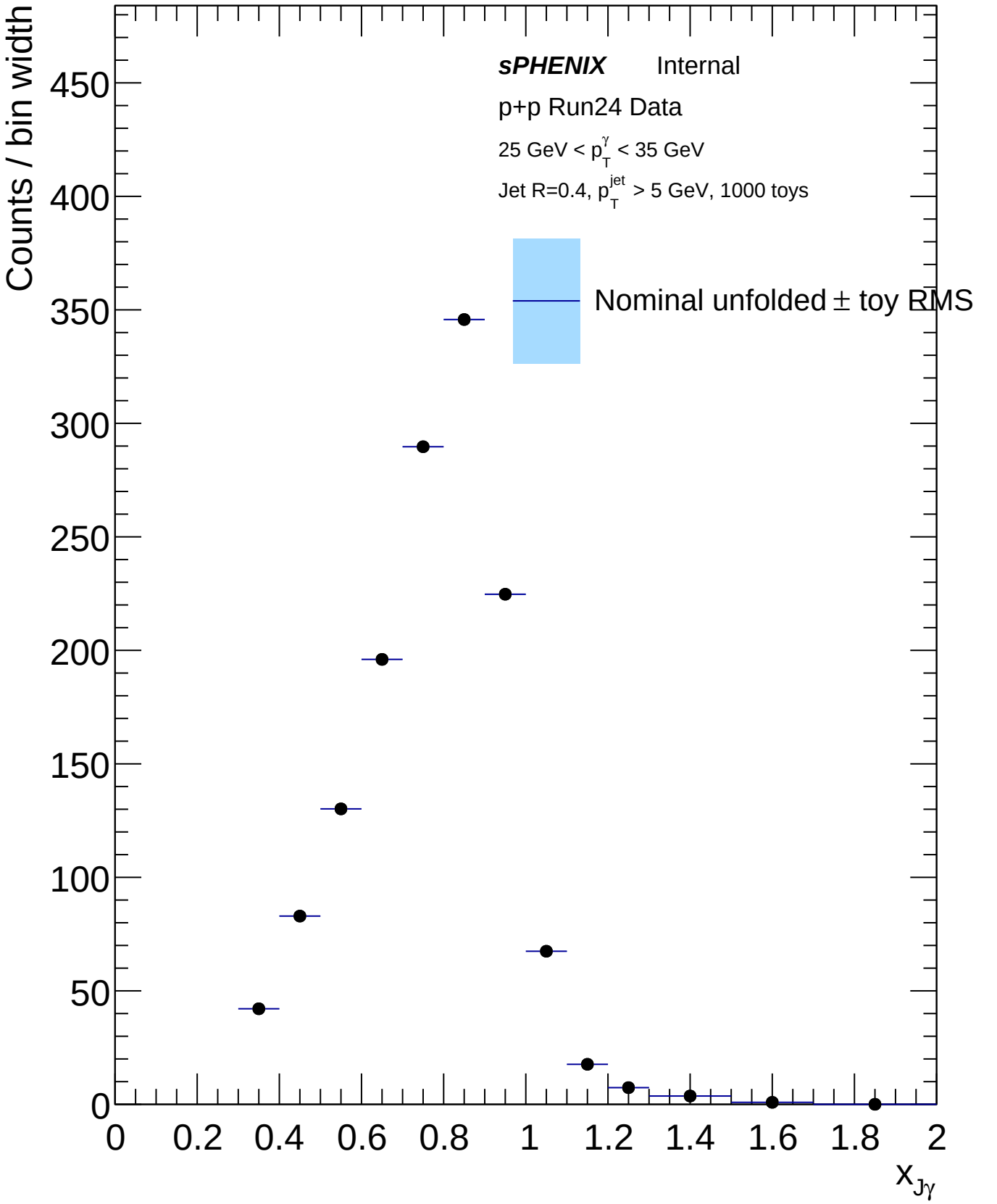

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Nominal unfolded
50 of 1000 response-matrix toys
Counts / bin width
6000
5000
4000
3000
2000
1000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Nominal unfolded
50 of 1000 response-matrix toys
Counts / bin width
0
200
400
600
800
1000
1200
1400
1600
1800
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV
Nominal unfolded
50 of 1000 response-matrix toys
Counts / bin width
0
50
100
150
200
250
300
350
400
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Jet R=0.4, p_T^{jet} > 5 GeV, 1000 toys
Nominal unfolded ± toy RMS
Counts / bin width
$x_{J\gamma}$
0
1000
2000
3000
4000
5000
6000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Jet R=0.4, p_T^{jet} > 5 GeV, 1000 toys
Nominal unfolded ± toy RMS
Counts / bin width
$x_{J\gamma}$
0
200
400
600
800
1000
1200
1400
1600
1800
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^{jet} > 5 GeV, 1000 toys
Nominal unfolded ± toy RMS
Counts / bin width
$x_{J\gamma}$
0
50
100
150
200
250
300
350
400
450
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2